SMART INDIA HACKATHON 2026

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- **Problem Statement ID** – SIH26037
- **Problem Statement Title -** Adaptive Path Planning and Collision Avoidance for Autonomous Vehicles on Unstructured Indian Roads
- **Theme-** Smart Vehicles
- **PS Category-** Software
- **Team ID -**
- **Team Name -** Neuralnauts

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IDEA TITLE

Proposed Solution

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What the Model thinks...

01 LiDAR and Cameras

02 YOLO Detection

03 Object Classification

04 Object Tracking

05 Risk Assessment

- A closed-loop adaptive autonomy stack for real-time mixed-traffic navigation on ill-structured pathways, ESPECIALLY, for rural indian roads.
- Development of a model which is highly perceptive of pattern recognition, prediction and planning by using sensors and 3D mapping. Further, deploying the plan by using different mechanics like stateflow, simulink and roadrunner.
- Our approach is designed around uncertainity, not just to accomodate perfect roads. It focuses on lane-agnostic planning, uncertainity awareness, risk driven replanning and Indian traffic adaptation.
- Our algorithm is developed on unsupervised learning model. It is capable of improving over time by assessing inputs from repeated runs and minimizing discomfort.

Object Classification

Object: Motorcycle
Type: Dynamic
Distance: 8.2 m
Relative motion: Approaching
TTC: 2.8 s
Risk: HIGH

Object: Pothole
Type: Static
Position: Left side
Road width: 4.2 m
Risk: MEDIUM

UNKNOWN OBJECT ↓

Increased uncertainty ↓

Reduce speed ↓

Increase safety margin ↓

Re-evaluate Approach ↓

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TECHNICAL APPROACH

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TTC — Time to Collision

$$TTC = \frac{D}{V_{relative}}$$

where:
D = distance to obstacle
Vrelative = relative speed

TTC > 30 sec → SAFE
=15–25 s → CAUTION
=7–15 sec → HIGH RISK
< 7 sec → EMERGENCY

- The hardware used in this project are LiDAR for observing 3-D gemotery, radar for detecting range and velocity and Camera for providing our system with computer vision.
- The framework we made focused on creating a lane-based system which can detect changes and blocks which allow it to further take decisions by predicting the optimal path to be followed.
- Our framework also accounts for real-time changes and acts after immediately processing the new data. The framework also accounts for irregular behaviour, where it uses context cues such as the road's edge and crowd density. We utilised Python-3 to bring our projected plan to actuation.

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FEASIBILITY AND VIABILITY

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Stage	Input	Processing	Output
1. Sense	Road & terrain	Data acquisition	Environment
2. Understand	Raw data	Preprocessing +	Road/terrain
3. Predict	Environment	AI/ML	Risk / route
4. Plan	Predicted conditions	Optimization	Optimal route
5. Adapt	New road conditions	Real-time replanning	Updated route
6. Act	Updated route	ADAS interface	Navigation / alerts
7. Learn	Real-world feedback	Retraining	Improved model

- The development and deployment of the model is extremely feasible, as the simulation and coding aspect can easily be done in MATLAB/Simulink, CARLA, ROS/ Gazebo, or Python

- The real life application of our system targets traversing varying terrain and aspects of every indian area.

- Trained and tested model on data specifically from unstructured Indian roads and using optimized algorithms to achieve real-time replanning.

- This model can be optimized for high-speed, high-risk maneuvers going further into real rough terrain.

- Since our model has been programmed on majorly free resources, such as python and MATLAB our system can be easily integrated into existing primitive ADAS systems

Component	Technology	Feasibility	Reason
Terrain & road-data processing	Python / MATLAB	🟢 High	Mature libraries and existing workflows
Simulation	MATLAB/Simulink, CARLA, Gazebo	🟢 High	Suitable simulation environments available
AI/ML model training	Python	🟢 High	Extensive ML ecosystem
Real-time replanning	Optimized algorithms	🟢 High	Can be designed for low-latency inference
Indian-road deployment	ADAS integration	🟡 Medium–High	Requires vehicle-specific
Rough-terrain operation	Sensors + perception + planning	🟡 Medium–High	Requires extensive real-world validation
Large-scale deployment	Cloud/edge + ADAS	🟡 Medium–High	Depends on hardware and integration

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IMPACT AND BENEFITS

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“FROM NAVIGATING ROADS TO UNDERSTANDING THEM”

- An autonomous driving system for unstructured indian roads can provide the Indian citizen with a new tool in their arsenal for emergencies as well as leisure.
- A trained version could anonymously record detected potholes, damaged roads and recurring obstacles and generate a Road Health Map (RHM) turning vehicles into road-condition data collectors.
- Emergency and public-service applications of our model can, for instance, reduce patient risk in ambulances, rdeuce article damge in delivery vehicles, help in road inspections and damage control after disasters.
- Our model provides an opportunity to those with disabilities to experience more freedom and control over their own lives.
- This system predicts that its planned trajectory will become unsafe if some obstacle arises and dynamically generates a safer alternative trajectory potentially preventing fatal accidents.

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RESEARCH AND REFERENCES

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Details / Links of the reference and research work :

- Survey of Autonomous Vehicles' Collision Avoidance Algorithms
- Evaluation of Path-Planning Algorithms
- Obstacle Straddling and Avoidance Integrated Path Planning and Tracking
- OpenCV
- Matplotlib
- NumPy